ABHIMANYU KK

LEAD EMBEDDED ENGINEER

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SUMMARY

Embedded Technical Lead with 7+ years of experience architecting and delivering high-performance embedded systems across complex, mission-critical environments. Proven leader driving projects from concept to production — owning system architecture, requirement analysis, technical roadmap, and cross-functional execution. Experienced in customer-facing engagements, onsite solution architecture discussions, and translating business requirements into scalable, production-ready designs.

Expert in creating comprehensive system architectures, writing Software Requirement Specifications (SRS), and reviewing Product Requirement Documents (PRD) to ensure technical feasibility, risk mitigation, and delivery alignment. Strong background in hardware–software co-design, including hardware architecture, schematic design, PCB layout, board bring-up, and system validation.

Deep technical expertise in **C, C++**, and **Python**, with advanced experience in **Embedded Linux**, including **Linux device driver** development, **kernel configuration** and customisation using the **Yocto Project**, custom **Linux** OS development, driver integration, OS optimisation, and performance tuning. Strong experience with **RTOS** environments and low-level hardware interfacing.

Skilled in modern embedded UI development using **Qt/QML and LVGL**, applying MVC design principles to build scalable, maintainable, and production-grade embedded systems.

SKILLS

Languages & Frameworks

- Embedded C/C++
- Python
- QML
- LVGL
- PyQt
- C#
- JS & TS
- Node JS
- React
- HTML
- CSS
- Rust (Basics)

Communication Protocols

- ModBus
- HiSLIP
- VXI-11
- TCP/IP
- USBTMC
- GPIB / IEEE488
- BLE
- WiFi
- ZigBee
- UART
- I2C
- SPI
- QSPI
- OSPI

Others

- Embedded Linux
- Device Drivers
- Yocto Project
- Zephyr
- GitHub
- GitLab
- Tortoise SVN
- Bit Bucket
- Docker
- CI/CD
- Jenkins
- Rally
- Aha
- KiCAD
- Altium
- IAR Workbench
- VS Code Extension dev.
- CMakeLists
- MakeFile

PROFESSIONAL EXPERIENCE

Lead Embedded Engineer, Senior Embedded Systems Engineer, Embedded Engineer

Utthunga Technologies | Bangalore | May 2021 - Present

- Led the design and development of embedded software, hardware, web server, and desktop application projects, delivering high-performance, reliable, and scalable solutions.
- Owned end-to-end project execution, managing timelines, resources, risks, and **cross-functional coordination** while leading a team of junior developers and test engineers.
- **Architected** the embedded system solution for the redesign of a multi-functional calibrator, defining system-level architecture, hardware–software partitioning, communication protocols, and scalability strategy.
- Designed cybersecurity requirements aligned with **CFR 21 Part 11** and **EU regulatory compliance standards**, incorporating secure authentication, audit trails, data integrity controls, and secure communication mechanisms.
- Conducted detailed risk assessments (technical and regulatory), identifying failure modes, mitigation strategies, and compliance gaps early in the design lifecycle.
- Reviewed Product Requirement Documents (PRD) and authored comprehensive Software Requirement Specifications (SRS), ensuring traceability, feasibility, and alignment with business and regulatory objectives.
- Led onsite customer discussions, gathering product requirements and translating them into **technical architecture**, **execution plans**, **delivery milestones**, and **commercial proposals**.
- Developed detailed project proposals including cost estimation, resource planning, risk evaluation, and delivery roadmaps aligned with strategic business goals.
- Executed Proof of Concept (POC) studies for platform selection (e.g., **Zephyr vs Embedded Linux**), benchmarking performance, scalability, real-time capabilities, and long-term maintainability.
- Performed EVK bring-up and board initialisation with custom OS builds, including boot-loader configuration, kernel customisation, driver integration, and hardware validation.
- Performed product certification and compliance testing, ensuring adherence to **LXI (LAN eXtensions for Instrumentation)** standards and regulatory requirements.
- Architected development workflows and maintained robust version control systems, enabling seamless collaboration and structured release management.
- Conducted rigorous source code reviews and design rule checks, enforcing best practices to ensure maintainable, secure, and production-grade code quality.
- Managed project tracking using structured project management tools, facilitating agile ceremonies and ensuring transparent stakeholder communication.
- Led technical recruitment efforts, conducting interviews to evaluate embedded systems expertise, architectural thinking, and problem-solving capabilities.
- Provided high-level technical support across multiple programs, resolving complex system issues and optimising performance bottlenecks.
- Built and mentored a specialised technical team, fostering capability growth and delivering projects successfully in a Reporting Manager capacity.
- Designed and implemented **bare-metal state machine architectures for microcontrollers**, developing deterministic, resource-optimised firmware for time-critical applications.

Technology Associate

Toutche Electric | Mysore | February 2021 - May 2021

- Developed embedded software for **electric bikes**, optimising performance and reliability.
- Debugged hardware issues, focusing on power-related faults to ensure system stability and efficiency.
- Performed PID tuning for **BLDC motors**, enhancing motor control precision and responsiveness.
- Collaborated with hardware and firmware teams to refine system integration and troubleshoot critical issues.
- Conducted system testing and validation, ensuring compliance with performance and safety standards.

Embedded System Developer

ICPro Solutions Pvt. Ltd. | Bangalore | January 2019 - February 2021

- Designed and developed **Industrial IoT** (IIoT) projects, integrating embedded systems with cloud and edge computing solutions.
- Developed end-to-end IIoT product development, from concept and prototyping to deployment and maintenance.
- Worked with various wireless communication protocols, including **BLE, Wi-Fi, ZigBee, LoRa, and NB-IoT**, optimising connectivity for industrial applications.
- Designed schematics and developed **multi-layer PCBs**, ensuring high-performance and EMI-compliant hardware designs.
- Developed **Android applications** for BLE-based projects, enabling seamless device communication and remote monitoring.
- Created desktop software solutions for embedded hardware configuration, diagnostics, and data visualisation.
- Integrated embedded systems with cloud platforms (AWS IoT, Azure IoT, Google Cloud IoT) for real-time monitoring and analytics.
- Implemented security features for IoT devices, including encryption, secure boot, and OTA firmware updates.
- Conducted hardware debugging and testing, ensuring robust and reliable operation in industrial environments.
- Collaborated with cross-functional teams, including firmware, hardware, and software engineers, to deliver high-quality products.

Internship Trainee

Keltron Advanced Studies | Calicut | June 2019 - January 2019

- Developed embedded software for traffic light systems, ensuring reliable real-time operation and synchronisation.
- Designed and implemented control algorithms to optimise traffic flow and integrate adaptive signalling.
- Developed compact, small-form-factor PCB designs for space-constrained embedded applications.
- Integrated wireless communication protocols for remote monitoring and control.
- Collaborated with urban planning and infrastructure teams to integrate smart traffic solutions with IoT-based monitoring.

PROJECTS

Multi-Function Calibrator

Technical Lead - Embedded

- Led end-to-end system definition from **PRD review to SRS authoring**, translating business and regulatory requirements into structured technical specifications.
- Conducted **platform evaluation and Proof-of-Concept (PoC)** studies to select the optimal SoC, OS, and hardware architecture for performance-critical instrumentation.
- Developed and customised a performance-optimised **Embedded Linux OS** using **Yocto**, including kernel configuration, driver integration, and OS performance tuning for LVGL graphical benchmarking.
- Developed a **custom Linux OS** for the NXP i.MX8ULP EVK platform, including:
  - Kernel configuration and customisation
  - **Device Tree** selection and modification for display integration
  - Display subsystem enablement and validation
  - Power management and sleep mode configuration/optimisation
  - Creation of custom application recipes
  - Development of a dedicated Yocto **meta-layer** for product-specific components
  - Build system customisation and image optimisation
- Designed the **high-level system architecture**, covering boot flow, secure update mechanisms, communication subsystems, and layered software design.
- Defined and implemented a **secure boot and A/B software update strategy**, ensuring authenticity, integrity, rollback protection, and field reliability.

- Architected modular **protocol subsystems** for instrument communication and external control interfaces.
- Designed a layered software architecture (HAL, BSP, middleware, application) to ensure scalability, testability, and maintainability.
- Established the **development methodology**, including coding standards, review workflows, CI/CD pipelines, and release management.
- Defined the **system test strategy**, covering unit, integration, hardware-in-loop (HIL), regression, and compliance validation.
- Implemented a structured **DevOps approach**, including containerised build environments, automated pipelines, and reproducible firmware builds.
- Designed and institutionalised a **Git branching strategy** tailored for a multi-team embedded product lifecycle.
- Defined and incorporated **cybersecurity requirements**, including secure boot, encrypted storage, secure communication channels, and vulnerability management.
- Led compliance alignment for **EU Cyber Resilience Act (CRA)** requirements, integrating secure development lifecycle practices.
- Defined technical approach for **CFR 21 Part 11 compliance**, focusing on audit trails, electronic records integrity, and system validation strategy.
- Developed and customised a **performance-optimised embedded Linux OS** for analysing and benchmarking LVGL graphical performance.
- Mentored cross-functional teams across firmware, application, QA, and DevOps to ensure architecture consistency and regulatory readiness.

Pressure Automated and Control Equipment

Scrum Team Lead | Sr. Embedded Developer

- Designed the software architecture for all required external interface protocols for a novel embedded product.
- Developed a C++ application on the **Torizon** Linux platform, ensuring high performance and modularity.
- Dockerized the entire application, streamlining deployment and cross-platform compatibility.
- Implemented **SCPI** communication over multiple protocols, including USB Serial, USB VCP, USB TMC, Raw Socket, HiSLIP, VXI-11, GPIB, and HTTP(S).
- Led an **8-member team**, including developers and testers, ensuring timely delivery of high-quality products.
- Developed **protocol stacks** in C++, optimising efficiency and maintainability.
- Designed and developed a custom **USBTMC Linux device driver** for this product, featuring:
  - **Signal interrupt** handling
  - **IOCTL**-based state machine architecture
  - Robust data transfer management
  - Tight integration with the SCPI application layer
- Developed a C++ library stack for **IEEE 488**-based National Instruments chips, ensuring seamless instrument communication.
- Built a Windows-based configuration tool in C# (Visual Studio) for configuring SCPI commands and auto-generating code.
- Maintained a GitHub Team repository, conducting thorough code reviews and enforcing best practices.
- Created UI/UX designs in Figma for web pages, ensuring a modern, intuitive, and user-friendly interface.
- Developed a web server using **Node.js, React, and Vite**, providing a fast and interactive web interface for the product.
- Implemented a static web front-end with React and Vite, enhancing usability and performance.
- Designed and tested the product for **LXI compliance**, ensuring compatibility with industry standards.
- Developed a **QML**-based UI, providing a modern, responsive, and scalable graphical interface.
- Designed and implemented a sound library using **ALSA**, enabling high-quality audio processing and playback.
- Integrated an **mDNS** (Multicast DNS) library, enabling seamless network discovery and connectivity, using Avahi Library.
- Implemented CI/CD pipelines using **GitHub Actions**, automating build, test, and deployment workflows for embedded software projects.

- Developed custom GitHub Actions scripts to handle automated firmware testing, **Docker** image builds, and deployment to target devices.
- Ensured product compliance with the latest cybersecurity requirements, including secure communication protocols and system hardening.
- Conducted extensive product testing and certification, meeting industry and regulatory standards.

Functional Safety

Sr. Firmware Developer

- Designed and developed firmware in Embedded C for the **Atmel SAMD21** micro-controller, ensuring optimised and efficient operation.
- Developed identical firmware for multiple redundant boards (legs) mounted on a single baseboard, enabling fault tolerance.
- Implemented automatic redundancy switching, allowing the system to detect and shift the active state based on the status of neighbouring legs.
- Configured the SAMD21 micro-controller to operate at its maximum clock frequency, ensuring optimal performance.
- Designed three operational modes: Active, Redundant, and Fail, enhancing system reliability and fault recovery.
- Configured high-speed **SPI** and **UART** for maximum data throughput, improving communication efficiency.
- Developed and optimised **ADC** and **DAC** functionality, ensuring precise analog signal acquisition and high-resolution digital-to-analog conversion.
- Configured the ADC for high-speed sampling with accurate reference voltage calibration.
- Implemented **DMA**-based ADC readings, reducing CPU load and enabling real-time signal processing.
- Configured the DAC for smooth waveform generation and precise voltage output control.
- Worked with **IAR Embedded Workbench**, leveraging advanced debugging and optimisation tools.

ASIC firmware regression testing

Team Lead

- Designed a firmware regression test platform in **Python** to execute lakhs of test cases for a novel high-resolution ADC ASIC.
- Developed a **test jig** to integrate automated testing into the CI/CD pipeline, ensuring seamless validation.
- Implemented Jenkins scripts for CI/CD automation, enabling continuous testing and deployment.
- Proposed an innovative test architecture on top of the Python Unit Test Library, significantly reducing the effort required for writing and automating massive test case execution.
- Automated test case generation, improving test coverage and efficiency while minimising manual effort.
- Led a 2-member team, ensuring effective test strategy execution, optimising resource allocation, and meeting project deadlines.

Electric Bike

Firmware Developer

- Developed embedded software in C for Infineon micro-controllers, ensuring efficient motor control and system reliability.
- Worked in Infineon Dev Studio, utilising its toolchain for development, debugging, and optimisation.
- Gained expertise in BLDC motor fundamentals, including its operational characteristics and control strategies.
- Implemented motor control features, including braking, acceleration, speed control, over-speed regulation, and traction control, ensuring smooth and safe operation.
- Established CAN communication between the micro-controller and the CAN display, enabling real-time monitoring of speed, torque, and system status.
- Developed and fine-tuned a PID control algorithm to achieve optimal torque and speed regulation, enhancing motor efficiency and performance.

- Optimised firmware for real-time motor control, improving response time and minimising latency in system feedback loops.
- Implemented safety mechanisms, including over-current, thermal protection, and fail-safe modes, ensuring system robustness.

Wireless IO Module

Embedded Developer

- Designed and developed schematic and PCB layouts for an IO module based on STM32 microcontroller and ZigBee, ensuring optimal performance and reliability.
- Developed firmware in Embedded C for the STM32 MCU, implementing low-level drivers for UART, SPI, and I2C to enable seamless peripheral communication.
- Configured ZigBee for mesh networking, allowing unlimited range within an industrial warehouse for robust wireless connectivity.
- Implemented real-time data acquisition and control in STM32, ensuring precise IO operations.
- Designed and implemented boot loader functionality, enabling OTA (Over-the-Air) firmware updates for field-upgradable devices.
- Created a Windows-based serial configuration tool in Visual Studio, allowing users to configure module settings and update firmware.
- Designed and developed both hardware and firmware for the gateway, using a Raspberry Pi Compute Module for high-performance processing and scalability.
- Conducted hardware validation and debugging, using oscilloscopes, logic analysers, and protocol analysers to ensure system stability.

Vibration Sensor

Embedded Developer

- Developed firmware for Nordic Semiconductor BLE SoC, enabling real-time data acquisition from an STM-based IMU (Accelerometer, Gyroscope, Magnetometer) and transmission over Bluetooth Low Energy (BLE).
- Implemented BLE communication protocols, ensuring efficient and low-latency data transfer between the sensor device and an Android application.
- Designed custom BLE services, including:
  - BLE UART for bidirectional serial communication between the device and mobile/PC applications.
  - BLE Beacon functionality, enabling proximity-based detection and advertisement of sensor data.
- Designed a custom BLE communication protocol, ensuring robust and error-free data exchange between the device and Android app.
- Implemented Over-the-Air (OTA) firmware updates, allowing remote firmware upgrades without physical access.
- Optimised BLE power management, reducing energy consumption and extending battery life for continuous operation.
- Developed low-level drivers for the IMU sensor, optimising sampling rates, filtering, and event detection.
- Performed extensive testing and validation, ensuring accuracy, reliability, and compliance with BLE standards.

Torque sensor module

Embedded Developer

- Developed firmware in Embedded C for precise strain gauge measurement, ensuring accurate torque sensing.
- Designed and implemented communication between the microcontroller and the ADC (ADS1115), a 32-bit resolution chip, for high-precision analog-to-digital conversion.
- Configured and optimised ADC registers, fine-tuning gain, sampling rate, and resolution to achieve high accuracy.
- Developed a calibration algorithm to eliminate offsets and improve the accuracy of torque measurements.
- Designed a real-time data transmission system over Infrared (IR) LED, enabling a Line-of-Sight gateway to receive strain gauge readings wirelessly.
- Implemented filtering techniques (e.g., moving average, low-pass filtering) to reduce noise and improve signal quality.

- Developed low-power firmware, optimising ADC sampling and communication to extend battery life.
- Performed extensive testing and validation, ensuring repeatability and reliability of torque measurements.

AIS140 Vehicle Location Tracking module

Freelancer | Hardware engineer

- Designed schematics for a Vehicle Location Tracking (VLT) system, ensuring compliance with AIS 140 standards.
- Developed multi-layer PCBs using KiCAD, optimising signal integrity, power distribution, and EMI/EMC performance.
- Integrated GPS, GSM, IMU, and RTC modules, enabling real-time vehicle tracking and telemetry.
- Designed emergency button functionality, ensuring instant alert transmission in compliance with AIS 140 safety regulations.
- Implemented power management strategies, optimising power consumption for continuous operation.
- Conducted PCB validation and debugging, using oscilloscopes, logic analysers, and network analysers.
- Performed hardware-level testing, ensuring the system met automotive-grade reliability and durability standards.
- Led the AIS 140 certification process, coordinating with regulatory bodies to meet mandatory compliance requirements.

Knee Analyser

Freelancer | Android developer

- Developed an Android application to analyse knee movement and posture using two identical IMU-BLE sensors.
- Implemented real-time BLE communication, ensuring seamless data acquisition from dual IMU sensors.
- Designed and developed algorithms to calculate the angle between two IMU sensors, enabling accurate knee motion tracking.
- Analysed and classified user postures (Sit, Stand, Sleep) based on real-time angle computations.
- Implemented data visualisation for knee motion analysis, allowing users to monitor their posture changes over time.
- Performed real-world testing and validation, ensuring high accuracy and reliability in movement classification.

Website

Freelancer | Website developer

- Designed and developed a website for a charity organisation, ensuring a user-friendly and visually appealing interface.
- Registered the domain and managed web hosting, setting up a reliable and secure online presence at www.racharity.org.in.
- Implemented responsive web design, ensuring seamless accessibility across desktops, tablets, and mobile devices.
- Developed the website using modern web technologies (e.g., React, JS).
- Integrated donation functionality, enabling secure online transactions to support fundraising efforts.
- Optimised website performance improving visibility and loading speed.
- Provided ongoing maintenance and updates, keeping the site functional and up to date.

UI/UX

Freelancer | UI/UX developer

- Designed UI/UX in Figma for various mobile applications, ensuring intuitive and user-friendly interfaces.
- Developed prototypes and wireframes for applications, including:
- Restaurant Management System – Streamlined table reservations, order management, and payment workflows.
- Delivery Application – Optimised order tracking, real-time notifications, and delivery route visualisation.
- User Management System – Designed role-based access, account management, and onboarding flows.
- Created static design layouts with well-structured navigation bars for seamless user interaction.
- Implemented simple flow animations to enhance user experience and visual appeal.
- Focused on responsive and accessible design, ensuring optimal usability across various screen sizes.

- Collaborated with developers, providing design assets and style guides for smooth implementation.

ACADEMICS

International Institute of Management and Technical (IIMT) Studies

Professional Graduate programme in Electronics and Telecommunication (2022 to 2024)

Malabar Institute of Technology (MIT)

Bachelor of Technology in Electronics and Communication - Course completion (2014 to 2018)

AWARDS & CERTIFICATIONS

Best Employee of the Year 2022

Utthunga Technologies

Accomplishments in Kick Start 2021

Google Kick Start 2021 - Ranked 6372nd across the globe

Embedded Linux Training

Utthunga Technologies

CONTACT

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	Valid Till: October 2027